

Equilibrium Operators, Variational Inequalities and Welfare:

What Is and Is Not a Theorem

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Abstract

Research line EQO001 asked whether a formal relationship between equilibrium operators, variational inequalities and welfare criteria could yield results beyond the existing ones; its public record calls the formulation reviewed a combination of known components without an independent theorem. This note records that assessment. With equilibria the solutions of $VI(F, K)$ on a closed convex set K and welfare a C^1 functional on K , we prove the classical facts on which any such relationship rests: potentials, monotonicity, the equilibrium–welfare gap (Pigou, 4/3) and the first-order coincidence whose repair is marginal-cost pricing. The one small addition concerns the cone $\Lambda(x^*)$ of welfare weights for which a Nash equilibrium maximises $\sum_i \lambda_i u_i$: a closed convex cone inside a first-order cone $\Lambda_1(x^*)$ built from $F(x^*)$, the normal cone and the marginal externalities, with equality under joint concavity; (F, K) does not determine it. Exact examples show that $\Lambda(x^*)$ need not be polyhedral (three players; three commodities with affine costs) and that $\Lambda \neq \Lambda_1$ for two commodities; deciding $\lambda \in \Lambda(x^*)$ is co-NP-complete, even for strongly monotone quadratic games. Numerical checks on 400 random monotone games confirm equality in every tested direction of every concave instance. No new equilibrium concept is offered.

1 Introduction

The variational-inequality formulation of equilibrium is one of the unifying ideas of applied mathematics: a Wardrop traffic equilibrium [34], a Nash equilibrium of a concave game [23, 27] and a Walrasian or spatial price equilibrium can all be written as: find x^* in a closed convex set K with $\langle F(x^*), x - x^* \rangle \geq 0$ for all $x \in K$ [6, 10, 12, 14, 17, 22]. Welfare is an optimisation problem over the same K , and the temptation that motivated research line EQO001 is obvious: to look for a “formal relationship” between the inequality and the optimisation problem from which welfare statements about equilibria would follow.

The public record of the line is candid: the formulation reviewed combined known components without establishing an independent theorem, and the line was paused. This note discharges the line’s objective in the only way that seems honest to us: by writing down, with complete proofs where they are short, what is a theorem in this area, what is a definition in disguise, and what remains open. We add one small result on the set of welfare weights that make a given equilibrium optimal; Remark 5.2 states which parts of it are standard and which are, as far as we could check, only repackaged here.

Contributions and scope. Section 2 fixes the setting; Section 3 proves the classical results; Section 4 gives the first-order characterisation of welfare-optimal equilibria and marginal-cost pricing; Section 5 contains the welfare-weight cone (Theorem 5.1), the impossibility statement

(Proposition 5.3), five exact examples, three of them negative (Examples 5.6, 5.7, 5.8), and the complexity of deciding membership (Proposition 5.9); Sections 6–7 report the numerical checks, what can be claimed, limitations and one open question. Everything is finite-dimensional and deterministic; every number in the text is produced by the scripts of Appendix A with fixed seeds.

2 Setting

Throughout, $K \subseteq \mathbb{R}^n$ is a nonempty closed convex set, $U \supseteq K$ an open set and $F: U \rightarrow \mathbb{R}^n$ a continuous map, the *equilibrium operator*. We write $\langle \cdot, \cdot \rangle$ for the Euclidean inner product.

Definition 2.1 (Variational inequality, normal cone). $\text{VI}(F, K)$ is the problem of finding $x^* \in K$ such that $\langle F(x^*), x - x^* \rangle \geq 0$ for all $x \in K$; its solution set is $\text{SOL}(F, K)$. The *normal cone* of K at $x \in K$ is $N_K(x) = \{v \in \mathbb{R}^n : \langle v, y - x \rangle \leq 0 \ \forall y \in K\}$, a closed convex cone, equal to $\{0\}$ when x is an interior point. By definition,

$$x^* \in \text{SOL}(F, K) \iff -F(x^*) \in N_K(x^*). \quad (1)$$

Definition 2.2 (Monotonicity). F is *monotone* on K if $\langle F(x) - F(y), x - y \rangle \geq 0$ for all $x, y \in K$; *strictly monotone* if the inequality is strict for $x \neq y$; μ -*strongly monotone* if $\langle F(x) - F(y), x - y \rangle \geq \mu \|x - y\|^2$ with $\mu > 0$; and L -*Lipschitz* if $\|F(x) - F(y)\| \leq L \|x - y\|$.

Definition 2.3 (Game). A *game* has N players, strategy sets $K_i \subseteq \mathbb{R}^{n_i}$ closed and convex, $K = K_1 \times \dots \times K_N$, and payoffs $u_i: U \rightarrow \mathbb{R}$ of class C^1 on an open $U \supseteq K$, with $u_i(\cdot, x_{-i})$ concave on K_i for every x_{-i} . A *Nash equilibrium* is $x^* \in K$ with $u_i(x^*) \geq u_i(y_i, x_{-i}^*)$ for all i and $y_i \in K_i$. The *pseudo-gradient* (equilibrium operator) is $F(x) = (-\nabla_{x_1} u_1(x), \dots, -\nabla_{x_N} u_N(x))$, where ∇_{x_i} is the gradient with respect to player i 's own variables. The *marginal externality* of player j on player $i \neq j$ at x is $\nabla_{x_j} u_i(x) \in \mathbb{R}^{n_i}$.

Lemma 2.4 (Nash equilibria are VI solutions). *In the setting of Definition 2.3, x^* is a Nash equilibrium if and only if $x^* \in \text{SOL}(F, K)$. [classical, 27; 10]*

Proof. For a concave differentiable function ϕ on a convex set C , z maximises ϕ over C iff $\langle \nabla \phi(z), y - z \rangle \leq 0$ for all $y \in C$: necessity because $t \mapsto \phi(z + t(y - z))$ is defined on $[0, 1]$ and has right derivative $\langle \nabla \phi(z), y - z \rangle$ at 0, which must be ≤ 0 ; sufficiency because $\phi(y) \leq \phi(z) + \langle \nabla \phi(z), y - z \rangle$. Apply this to $\phi = u_i(\cdot, x_{-i}^*)$ on K_i : x^* is Nash iff $\langle -F_i(x^*), y_i - x_i^* \rangle \leq 0$ for all i and $y_i \in K_i$. Summing over i gives $\langle F(x^*), y - x^* \rangle \geq 0$ for every $y \in K$; conversely, choosing $y = (y_i, x_{-i}^*)$ in the VI recovers the i -th inequality. $\square$

Definition 2.5 (Nonatomic routing). A network has edges E , commodities $k = 1, \dots, m$ with demands $d_k > 0$ and path sets P_k , $P = \bigsqcup_k P_k$. A path flow $f \in \mathbb{R}_+^P$ is feasible if $\sum_{p \in P_k} f_p = d_k$ for each k ; K is the set of feasible flows, a product of scaled simplices. Edge flows are $f_e = \sum_{p \ni e} f_p$, edge costs $c_e: \mathbb{R}_+ \rightarrow \mathbb{R}_+$ are continuous and nondecreasing, path costs are $c_p(f) = \sum_{e \in p} c_e(f_e)$, and the equilibrium operator is $F(f) = (c_p(f))_{p \in P}$. A *Wardrop equilibrium* is a feasible f^* such that, for every k and $p, q \in P_k$ with $f_p^* > 0$, $c_p(f^*) \leq c_q(f^*)$. The *social cost* is $C(f) = \sum_p f_p c_p(f) = \sum_e f_e c_e(f_e)$, and welfare is $W = -C$.

Lemma 2.6 (Normal cone of a face of a simplex). *Let $\Delta = \{x \in \mathbb{R}_+^P : \sum_{p \in P_k} x_p = d_k, \ k = 1, \dots, m\}$ and $x \in \Delta$ with support $S = \{p : x_p > 0\}$. Then*

$$N_\Delta(x) = \{v : \exists \gamma \in \mathbb{R}^m \text{ with } v_p = \gamma_k \text{ for } p \in S \cap P_k, \ v_p \leq \gamma_k \text{ for } p \in P_k \setminus S\}.$$

This is the polar of the tangent cone $\{h : \sum_{p \in P_k} h_p = 0 \ \forall k, \ h_p \geq 0 \ (p \notin S)\}$, computed by Farkas' lemma [26].

Lemma 2.7 (Wardrop equilibria are VI solutions). f^* is a Wardrop equilibrium iff $f^* \in \text{SOL}(F, K)$ with $F = (c_p)_p$. [classical, 31; 6]

Proof. By (1) and Lemma 2.6, $f^* \in \text{SOL}(F, K)$ iff for each k there is γ_k with $c_p(f^*) = -\gamma_k$ on used paths of P_k and $c_p(f^*) \geq -\gamma_k$ on unused ones, which is Wardrop's condition. $\square$

Definition 2.8 (Welfare functionals). A welfare functional is a C^1 function $W: U \rightarrow \mathbb{R}$ to be maximised over K . For a game, the weighted welfare with weights $\lambda \in \mathbb{R}_+^N$ is $W_\lambda = \sum_i \lambda_i u_i$; $\lambda = \mathbf{1}$ is the utilitarian criterion. For routing, $W = -C$.

3 The classical results, with proofs

3.1 Potentials

A C^1 field F with symmetric Jacobian on an open convex U is a gradient (Poincaré's lemma): $P(x) = \int_0^1 \langle F(x_0 + t(x - x_0)), x - x_0 \rangle dt$ satisfies $\partial_i P(x) = \int_0^1 \frac{d}{dt} [t F_i(x_0 + t(x - x_0))] dt = F_i(x)$, by differentiation under the integral and $\partial_j F_i = \partial_i F_j$.

Theorem 3.1 (Equilibria of gradient operators). Suppose $F = \nabla P$ on $U \supseteq K$ with $P \in C^1(U)$. Then

$$\text{SOL}(F, K) = \{x \in K : \langle \nabla P(x), y - x \rangle \geq 0 \ \forall y \in K\},$$

the set of stationary points of P over K ; it contains $\arg \min_K P$, and equals it when P is convex on K . If P is strictly convex, $\text{SOL}(F, K)$ has at most one point. [classical: 2 for routing, 19 for potential games; proved here]

Proof. The first identity is Definition 2.1 with $F = \nabla P$. If x minimises P over K then, as in the proof of Lemma 2.4, the right derivative of $t \mapsto P(x + t(y - x))$ at 0 is nonnegative, which is the stationarity inequality. If P is convex and x is stationary, $P(y) \geq P(x) + \langle \nabla P(x), y - x \rangle \geq P(x)$ for all $y \in K$. If P is strictly convex its minimiser over the convex set K is unique. $\square$

Remark 3.2 (The three classical instances). (a) *Routing* [2]. For continuous c_e , $P(f) = \sum_e \int_0^{f_e} c_e(s) ds$ is C^1 with $\partial P / \partial f_p = \sum_{e \in p} c_e(f_e) = c_p(f)$ (fundamental theorem of calculus), convex when the c_e are nondecreasing; Wardrop equilibria minimise P , with unique edge flows when all c_e are strictly increasing. $P \neq C$: the potential integrates the cost, the social cost multiplies it (Section 3.3). (b) *Concave games* [27]. Lemma 2.4 and Theorem 3.4(a) give uniqueness under Rosen's diagonal strict concavity, which for unit weights is strict monotonicity of the pseudo-gradient. (c) *Potential games* [19]. A C^1 game has exact potential Π iff $F = -\nabla \Pi$; its equilibria are the stationary points of $-\Pi$ over K . The potential is bookkeeping, not welfare (Example 5.5(i)).

3.2 Monotonicity: uniqueness, existence and a solver

Lemma 3.3 (Projection). Let Π_K be the Euclidean projection onto K . Then $z = \Pi_K(y)$ iff $z \in K$ and $\langle y - z, x - z \rangle \leq 0$ for all $x \in K$; Π_K is nonexpansive; and for every $\gamma > 0$, $x \in \text{SOL}(F, K)$ iff $x = \Pi_K(x - \gamma F(x))$. [classical, 17, Chapter I]

Theorem 3.4 (Monotone operators). (a) If F is strictly monotone on K , $\text{SOL}(F, K)$ has at most one point. (b) If F is μ -strongly monotone and L -Lipschitz on K , $\text{SOL}(F, K)$ is a singleton $\{x^*\}$, and for $0 < \gamma < 2\mu/L^2$ the iteration $x_{t+1} = \Pi_K(x_t - \gamma F(x_t))$ satisfies $\|x_{t+1} - x^*\| \leq q \|x_t - x^*\|$ with $q = \sqrt{1 - 2\gamma\mu + \gamma^2 L^2} < 1$; for $\gamma = \mu/L^2$, $q = \sqrt{1 - \mu^2/L^2}$. [classical, 10, 17; proved here]

Proof. (a) If $x, y \in \text{SOL}(F, K)$ then $\langle F(x), y - x \rangle \geq 0$ and $\langle F(y), x - y \rangle \geq 0$; adding, $\langle F(x) - F(y), x - y \rangle \leq 0$, so $x = y$ by strict monotonicity. (b) Let $T(x) = \Pi_K(x - \gamma F(x))$. By nonexpansiveness, $\|Tx - Ty\|^2 \leq \|(x - y) - \gamma(F(x) - F(y))\|^2 = \|x - y\|^2 - 2\gamma \langle F(x) - F(y), x - y \rangle$

$y\rangle + \gamma^2\|F(x) - F(y)\|^2 \leq q^2\|x - y\|^2$, and $q < 1$ iff $\gamma < 2\mu/L^2$. Banach's fixed-point theorem on the complete set K gives a unique fixed point, which is the unique VI solution by Lemma 3.3, and the contraction estimate is the stated rate. $\square$

Remark 3.5. For merely continuous F on compact convex K , existence is the Hartman–Stampacchia theorem (14, Lemma 3.1), obtained from Brouwer's theorem applied to T ; we do not reproduce it. All solvers in Section 6 are the iteration of Theorem 3.4(b) with $\gamma = \mu/L^2$, so each computed equilibrium comes with a uniqueness proof and a residual certificate; the exception is Example 5.6 (E2c), where F is constant, $\gamma = 1$ is used and uniqueness follows from dominance.

3.3 Equilibrium is not welfare: Pigou and the price of anarchy

Example 3.6 (Pigou). Two parallel links, unit demand, $c_1 \equiv 1$ and $c_2(x) = x^d$, $d \geq 1$ [25]. Every user on link 2 pays $1 \leq c_1$, so $f^* = (0, 1)$ is a Wardrop equilibrium with $C(f^*) = 1$, and it is the only one: a feasible flow with $f_1 > 0$ has $f_2 < 1$, so its users on link 1 pay $1 > f_2^d$. The social cost of $(1 - x, x)$ is $1 - x + x^{d+1}$, minimised at $x = (d+1)^{-1/d}$ with value $1 - \frac{d}{d+1}(d+1)^{-1/d}$. For $d = 1$ the optimum is $(\frac{1}{2}, \frac{1}{2})$ with cost $\frac{3}{4}$ and ratio $\frac{4}{3}$; the ratio tends to infinity with d , since $(d+1)^{-1/d} \rightarrow 1$. The computed values are in `results/tables_traffic.md`.

Theorem 3.7 (Price of anarchy for affine costs). *If every $c_e(x) = a_e x + b_e$ with $a_e, b_e \geq 0$, then for every Wardrop equilibrium f and every feasible f' , $C(f) \leq \frac{4}{3}C(f')$. The constant is attained in Example 3.6 with $d = 1$. [classical, 29, see also 28; the proof below follows 5]*

Proof. By Lemma 2.7, $\sum_p c_p(f)(f'_p - f_p) \geq 0$; regrouping by edges, $C(f) = \sum_e c_e(f_e)f_e \leq \sum_e c_e(f_e)f'_e$. For each edge, $c_e(f_e)f'_e = c_e(f'_e)f'_e + a_e(f_e - f'_e)f'_e \leq c_e(f'_e)f'_e + \frac{1}{4}a_e f_e^2 \leq c_e(f'_e)f'_e + \frac{1}{4}c_e(f_e)f_e$, since $y \mapsto a_e(f_e - y)y$ is maximised at $y = f_e/2$ and $a_e f_e^2 \leq (a_e f_e + b_e)f_e$. Summing, $C(f) \leq C(f') + \frac{1}{4}C(f)$. $\square$

4 When an equilibrium is a welfare maximiser

Proposition 4.1 (Necessary and sufficient first-order conditions). *Let $x^* \in \text{SOL}(F, K)$ and let W be differentiable at x^* .*

- (a) *If $x^* \in \arg \max_K W$ then $\nabla W(x^*) \in N_K(x^*)$ and $-F(x^*) \in N_K(x^*)$. If W is concave on K , the converse holds: $x^* \in \arg \max_K W$ iff $\nabla W(x^*) \in N_K(x^*)$.*
- (b) *If x^* is an interior point of K , (a) forces $F(x^*) = 0 = \nabla W(x^*)$. If $N_K(x^*) = \mathbb{R}_+ \nu$ is a ray and $F(x^*) \neq 0$, (a) forces $\nabla W(x^*) = -\theta F(x^*)$ for some $\theta \geq 0$.*
- (c) *(Simplices.) If K is a product of simplices (Lemma 2.6), both $F(x^*)$ and $\nabla W(x^*)$ are constant on the support of x^* within each commodity (hence parallel there when both are non-zero), and each satisfies Wardrop's principle with respect to its own coordinates. On $K = \{x \geq 0 : Ax = b\}$ the same holds with “constant on the support” replaced by “equal on the support to an element of the row space of A ”.*
- (d) *If F is strictly monotone, W is strictly concave and both $\text{SOL}(F, K)$ and $\arg \max_K W$ are nonempty, then $\text{SOL}(F, K) = \arg \max_K W$ iff the unique equilibrium satisfies $\nabla W(x^*) \in N_K(x^*)$; otherwise the two sets are disjoint.*

[proved here; elementary]

Proof. (a) The first-order necessary condition for a maximum of a differentiable function over a convex set (proof of Lemma 2.4) is $\langle \nabla W(x^*), y - x^* \rangle \leq 0$ for all $y \in K$, i.e. $\nabla W(x^*) \in N_K(x^*)$;

the second membership is (1); sufficiency under concavity is the same argument as in Theorem 3.1. (b) For interior x^* , $N_K(x^*) = \{0\}$. If $N_K(x^*) = \mathbb{R}_+\nu$, write $-F(x^*) = \alpha\nu$ and $\nabla W(x^*) = \beta\nu$ with $\alpha, \beta \geq 0$; $F(x^*) \neq 0$ gives $\alpha > 0$ and $\theta = \beta/\alpha$. (c) Lemma 2.6 applied to $-F(x^*)$ and to $\nabla W(x^*)$; for $\{x \geq 0 : Ax = b\}$ the polar of the tangent cone is $\{A^\top w - v : v \geq 0, v_S = 0\}$ by Farkas' lemma. (d) Both sets are singletons by Theorem 3.4(a) and strict concavity; apply (a). $\square$

Proposition 4.1 is a statement about first-order conditions and nothing more: it says that an equilibrium can be a welfare optimum only if the two problems happen to share their KKT point, and it gives no mechanism producing that coincidence. The one mechanism that always works is to change the operator.

Corollary 4.2 (Marginal-cost pricing and the toll set). *Let $C \in C^1(U)$ be a cost to be minimised over K and define the toll field $\tau = \nabla C - F$. Then*

$$\text{SOL}(F + \tau, K) = \{x \in K : -\nabla C(x) \in N_K(x)\} \supseteq \arg \min_K C,$$

with equality when C is convex on K , and at most one point when C is strictly convex; the tolled operator is a gradient field with potential C . For separable routing costs with $c_e \in C^1$, $\tau_p(f) = \sum_{e \in p} f_e c'_e(f_e)$, the Pigou–Knight–Beckmann marginal-cost toll. More generally, for a target $\bar{x} \in K$ the set of constant toll vectors $t \in \mathbb{R}^n$ with $\bar{x} \in \text{SOL}(F + t, K)$ is the translated cone $-F(\bar{x}) - N_K(\bar{x})$, polyhedral when K is; choosing in it the toll closest to a reference vector is an inverse optimisation problem in the sense of Ahuja and Orlin [1], and for routing it is the toll-set description of Hearn and Ramana [15]. [classical, 2, 15, 18, 25; proved here]

Proof. $F + \tau = \nabla C$, so the first statement is Theorem 3.1 with $P = C$. For routing, $\partial C / \partial f_p = \sum_{e \in p} (c_e(f_e) + f_e c'_e(f_e))$, whence τ_p . Finally $\bar{x} \in \text{SOL}(F + t, K)$ iff $-F(\bar{x}) - t \in N_K(\bar{x})$ by (1). $\square$

5 The cone of welfare weights

Fix a game as in Definition 2.3 and a Nash equilibrium x^* . Write $u = (u_1, \dots, u_N) : K \rightarrow \mathbb{R}^N$ and $G(x^*) = Du(x^*) \in \mathbb{R}^{N \times n}$ for the Jacobian of the payoff vector, whose i -th row is $\nabla u_i(x^*)^\top$ (all variables, own and others'). Define

$$\Lambda(x^*) = \{\lambda \in \mathbb{R}_+^N : x^* \in \arg \max_K W_\lambda\}, \quad \Lambda_1(x^*) = \{\lambda \in \mathbb{R}_+^N : G(x^*)^\top \lambda \in N_K(x^*)\}.$$

For a set $D \subseteq \mathbb{R}^N$, $D^\circ = \{\lambda : \langle \lambda, d \rangle \leq 0 \ \forall d \in D\}$ is its polar cone.

Theorem 5.1 (Welfare-weight cone). (a) $\Lambda(x^*) = \mathbb{R}_+^N \cap \{u(x) - u(x^*) : x \in K\}^\circ$. In particular $\Lambda(x^*)$ is a closed convex cone.

(b) $\Lambda(x^*) \subseteq \Lambda_1(x^*)$, and $\Lambda_1(x^*)$ is a closed convex cone, polyhedral whenever K is polyhedral.

(c) If every u_i is concave on K (jointly in all variables), then $\Lambda(x^*) = \Lambda_1(x^*)$. More generally, equality holds whenever W_λ is concave on K for every $\lambda \in \Lambda_1(x^*)$.

(d) (Block form.) $\lambda \in \Lambda_1(x^*)$ iff for every player j

$$-\lambda_j F_j(x^*) + \sum_{i \neq j} \lambda_i \nabla_{x_j} u_i(x^*) \in N_{K_j}(x_j^*). \quad (2)$$

Since $-F_j(x^*) \in N_{K_j}(x_j^*)$ already, a sufficient condition is that the λ -weighted externality $\sum_{i \neq j} \lambda_i \nabla_{x_j} u_i(x^*)$ lies in $N_{K_j}(x_j^*)$ for every j ; when x_j^* is interior to K_j , (2) says exactly that this weighted externality vanishes.

(e) $\Lambda(x^*) \neq \{0\}$ implies that x^* is weakly Pareto efficient in K (no $x \in K$ with $u_i(x) > u_i(x^*)$ for all i); under the concavity of (c), the converse holds. Without any concavity, if x^* is locally weakly Pareto efficient then $\Lambda_1(x^*) \neq \{0\}$.

[proved here; (a) and (e) are the standard scalarisation argument of multi-objective optimisation [9, 11], and $\Lambda(x^*)$ is the cell at $u(x^*)$ of the weight-set decomposition studied for linear problems by Benson and Sun [3]]

Proof. (a) $\lambda \in \Lambda(x^*)$ iff $\lambda \geq 0$ and $\langle \lambda, u(x) - u(x^*) \rangle \leq 0$ for all $x \in K$, which is the stated intersection; polar cones and the orthant are closed convex cones. (b) If x^* maximises the differentiable W_λ over K , the first-order condition gives $\nabla W_\lambda(x^*) = \sum_i \lambda_i \nabla u_i(x^*) = G(x^*)^\top \lambda \in N_K(x^*)$. $\Lambda_1(x^*)$ is the preimage of the closed convex cone $N_K(x^*)$ under a linear map, intersected with $\mathbb{R}_+^N$; when K is polyhedral so is $N_K(x^*)$, hence so is $\Lambda_1(x^*)$. (c) For $\lambda \geq 0$, W_λ is concave when the u_i are, and the first-order condition is then sufficient (Proposition 4.1(a)); the general statement is the same argument. (d) For a product set, $v \in N_K(x^*)$ iff $v_j \in N_{K_j}(x_j^*)$ for each j : taking $y \in K$ that differs from x^* in block j only shows necessity, and summing the block inequalities shows sufficiency. Block j of $G(x^*)^\top \lambda$ is $\sum_i \lambda_i \nabla_{x_j} u_i(x^*) = \lambda_j \nabla_{x_j} u_j(x^*) + \sum_{i \neq j} \lambda_i \nabla_{x_j} u_i(x^*)$ and $\nabla_{x_j} u_j(x^*) = -F_j(x^*)$. The sufficient condition follows because $N_{K_j}(x_j^*)$ is a convex cone containing $-F_j(x^*)$ by Lemma 2.4; for interior x_j^* , $N_{K_j}(x_j^*) = \{0\}$ forces $F_j(x^*) = 0$ and then (2) reads $\sum_{i \neq j} \lambda_i \nabla_{x_j} u_i(x^*) = 0$. (e) If $\lambda \in \Lambda(x^*) \setminus \{0\}$ and $u(x) > u(x^*)$ componentwise, then $\langle \lambda, u(x) \rangle > \langle \lambda, u(x^*) \rangle$, a contradiction. Conversely, under concavity the set $A = u(K) - \mathbb{R}_+^N$ is convex; weak efficiency means $u(x^*) + \varepsilon \mathbf{1} \notin A$ for every $\varepsilon > 0$, so $u(x^*)$ is a boundary point of A and a supporting hyperplane gives $\lambda \neq 0$ with $\langle \lambda, a \rangle \leq \langle \lambda, u(x^*) \rangle$ for all $a \in A$; since $u(x^*) - \mathbb{R}_+^N \subseteq A$, $\lambda \geq 0$, and $a = u(x)$ gives $\lambda \in \Lambda(x^*)$. Finally, if $\Lambda_1(x^*) = \{0\}$, the convex cone $\{sG(x^*)(x - x^*) : x \in K, s \geq 0\}$ meets the open orthant: otherwise a separating hyperplane (the Gordan–Motzkin alternative) gives $\lambda \geq 0$, $\lambda \neq 0$, with $\langle \lambda, G(x^*)(x - x^*) \rangle \leq 0$ on K , i.e. $\lambda \in \Lambda_1(x^*)$; and $G(x^*)(x - x^*) > 0$ gives $u(x^* + \tau(x - x^*)) > u(x^*)$ for small $\tau > 0$. $\square$

Remark 5.2 (What is standard and what is not). The scalarisation in (a) and (e) is textbook, and the set of weights under which a given point maximises the weighted sum is a standard object of multi-objective optimisation (the weight-set decomposition, polyhedral in the linear case [3]); Proposition 5.3 repackages the Nikaidô–Isoda gap function and is an instance of best-response equivalence [20]. What we did not find in the literature we consulted, and claim only as packaging, is the reading of (2): the cone is cut out by the equilibrium face (through $N_{K_j}(x_j^*)$ and $F_j(x^*)$, both determined by (F, K)) and by the marginal externalities $\nabla_{x_j} u_i(x^*)$, which are *not* determined by (F, K) . The last claim of (e) and the interior case of (d) give the first-order mechanism behind the theorem of Dubey [8] that, for interior equilibria and generically in the payoffs, Nash equilibria of smooth games are Pareto-inefficient: at an interior x^* , local weak efficiency forces some $\lambda \geq 0$, $\lambda \neq 0$, to annihilate every weighted externality; Example 5.5(i) is an instance.

Proposition 5.3 ((F, K) does not determine welfare). *Let the u_i be continuous on $K = \prod K_i$ with K_i compact, and let x^* be a Nash equilibrium. Define $v_i(x_{-i}) = \max_{y_i \in K_i} u_i(y_i, x_{-i})$ and $u'_i = u_i - v_i$. Then the game (u'_i) has the same best-response correspondences and the same Nash equilibria as (u_i) ; wherever $\nabla_{x_i} u_i$ exists, $\nabla_{x_i} u'_i = \nabla_{x_i} u_i$, so the two games have the same equilibrium operator; and x^* maximises every $W'_\lambda = \sum_i \lambda_i u'_i$, $\lambda \in \mathbb{R}_+^N$, over K : $\Lambda'(x^*) = \mathbb{R}_+^N$. Consequently welfare optimality of an equilibrium is not a property of (F, K) : every pair (F, K) that arises from a game is compatible with $\Lambda(x^*) = \mathbb{R}_+^N$; if $N \geq 2$ and some K_{-i} is not a singleton, it is also compatible with $e_i \notin \Lambda(x^*)$ (and the pair of Example 5.5(i) with $\Lambda(x^*) = \{0\}$). [proved here; $\sum_i (v_i - u_i)$ is the Nikaidô–Isoda gap function [24], and $u \mapsto u'$ a best-response equivalence [20]]*

Proof. v_i does not depend on x_i , so subtracting it changes neither $\arg \max_{y_i \in K_i} u_i(y_i, x_{-i})$ nor $\nabla_{x_i} u_i$. By construction $u'_i \leq 0$ on K , and $u'_i(x^*) = 0$ because x_i^* is a best response to x_{-i}^* ; hence $W'_\lambda \leq 0 = W'_\lambda(x^*)$ for every $\lambda \geq 0$. For the second claim, add to u_i the term $c \langle d, x_{-i} - x_{-i}^* \rangle$,

with d such that $\delta = \langle d, x_{-i} - x_{-i}^* \rangle > 0$ for some $x_{-i} \in K_{-i}$: neither F nor the equilibria change, the game stays in the class of Definition 2.3, and $x = (x_i^*, x_{-i})$ beats x^* for W_{e_i} once $c\delta > u_i(x^*) - u_i(x)$. $\square$

The modified game need not be C^1 , so $\Lambda'(x^*)$ is understood in the exact sense of Theorem 5.1(a). When best responses are unique, Danskin's theorem [7] gives $v_i \in C^1$ with $\nabla v_i(x_{-i}) = \nabla_{x_{-i}} u_i(b_i(x_{-i}), x_{-i})$; the modified game is then in the class of Definition 2.3 and has *zero marginal externalities at x^** , so $\Lambda_1'(x^*) = \mathbb{R}_+^N$ by Theorem 5.1(d), the first-order reason for the proposition.

Example 5.4 (Two players, exact cone). Let $K = [0, 1]^2$, $\alpha > 1$, $\beta_1, \beta_2 \geq 0$, and

$$u_1(x) = \alpha x_1 - \frac{1}{2}x_1^2 - \beta_1 x_2, \quad u_2(x) = \alpha x_2 - \frac{1}{2}x_2^2 - \beta_2 x_1.$$

Each player's action harms the other linearly. The equilibrium operator is $F(x) = (x_1 - \alpha, x_2 - \alpha)$, with Jacobian I , so F is 1-strongly monotone and 1-Lipschitz and does not depend on β ; the unique equilibrium is $x^* = (1, 1)$, at which $N_K(x^*) = \mathbb{R}_+^2$ and $F(x^*) = (1 - \alpha)\mathbf{1}$. Here $G(x^*) = \begin{pmatrix} \alpha-1 & -\beta_1 \\ -\beta_2 & \alpha-1 \end{pmatrix}$, so by (2)

$$\Lambda_1(x^*) = \{\lambda \geq 0 : (\alpha - 1)\lambda_1 \geq \beta_2 \lambda_2, (\alpha - 1)\lambda_2 \geq \beta_1 \lambda_1\}.$$

Both payoffs are jointly concave (Hessians $-\text{diag}(1, 0)$ and $-\text{diag}(0, 1)$), so $\Lambda(x^*) = \Lambda_1(x^*)$ by Theorem 5.1(c). Thus the cone is a wedge, non-trivial iff $\beta_1 \beta_2 \leq (\alpha - 1)^2$ and two-dimensional iff the inequality is strict; for $\alpha = 2$, $\beta_1 = \beta_2 = \frac{1}{2}$ it is $\{\frac{1}{2}\lambda_2 \leq \lambda_1 \leq 2\lambda_2\}$ (Figure 1): the corner equilibrium is welfare-optimal exactly when neither player is weighted more than twice the other. Games with different β have the same F and different cones: Proposition 5.3 in miniature.

Example 5.5 (Cournot duopoly). Let $u_i(q) = q_i(a - q_1 - q_2)$ on $K = [0, 1]^2$; $\partial^2 u_1 / \partial q_1^2 = -2 < 0$, so Lemma 2.4 applies, and $F(q) = (2q_1 + q_2 - a, q_1 + 2q_2 - a)$ has symmetric part $\begin{pmatrix} 2 & 1 \\ 1 & 2 \end{pmatrix} \succ 0$. (i) For $a = 1$ the equilibrium $q^* = (\frac{1}{3}, \frac{1}{3})$ is interior, $F(q^*) = 0$, and $\partial u_1 / \partial q_2 = -q_1^* \neq 0$: by Theorem 5.1(d) the interior condition fails for every $\lambda \neq 0$, so $\Lambda(q^*) = \{0\}$ (Dubey's inefficiency, Remark 5.2). (collusion at $(\frac{1}{4}, \frac{1}{4})$ gives each firm $\frac{1}{8} > \frac{1}{9}$). The game has the exact potential $\Pi(q) = a(q_1 + q_2) - q_1^2 - q_2^2 - q_1 q_2$, maximised at the equilibrium: a potential is not a welfare functional. (ii) For $a = 4$ the equilibrium is the corner $(1, 1)$, $F(q^*) = (-1, -1)$, $G(q^*) = \begin{pmatrix} 1 & -1 \\ -1 & 1 \end{pmatrix}$ and $\Lambda_1(q^*) = \{\lambda_1 = \lambda_2 \geq 0\}$, a ray. The payoffs are not jointly concave, so Theorem 5.1(c) is silent; but $W_{(1,1)}(q) = (q_1 + q_2)(4 - q_1 - q_2)$ is concave, so $\Lambda(q^*) = \Lambda_1(q^*)$ nevertheless. Concavity is sufficient, not necessary.

Example 5.6 (Three players: a non-polyhedral cone). Let $K = [0, 1]^3$ and

$$u_1 = x_1, \quad u_2 = x_2, \quad u_3 = x_3 - \frac{1}{2}x_1^2 + \frac{1}{4}x_1 + \frac{1}{2}x_1 x_2 - \frac{1}{2}x_2.$$

Each payoff is linear, hence concave, in the player's own variable, so the game is in the class of Definition 2.3; $F \equiv (-1, -1, -1)$, $x^* = (1, 1, 1)$ is the unique Nash equilibrium and $N_K(x^*) = \mathbb{R}_+^3$. Since $\nabla u_3(x^*) = (-\frac{1}{4}, 0, 1)$, $G(x^*)^\top \lambda = (\lambda_1 - \frac{1}{4}\lambda_3, \lambda_2, \lambda_3)$ and $\Lambda_1(x^*) = \{\lambda \geq 0 : \lambda_1 \geq \frac{1}{4}\lambda_3\}$. We compute $\Lambda(x^*)$ exactly. On $\{\lambda_3 = 0\}$, $W_\lambda = \lambda_1 x_1 + \lambda_2 x_2$ is maximised at x^* , so $\mathbb{R}_+^2 \times \{0\} \subseteq \Lambda(x^*)$; as $\Lambda(x^*)$ is a cone it suffices to fix $\lambda_3 = 1$. Then W_λ is increasing in x_3 , so $x_3 = 1$, and affine in x_2 , so the maximum over x_2 is attained at $x_2 \in \{0, 1\}$. For $x_2 = 1$, $W_\lambda = \frac{1}{2} + \lambda_2 + (\lambda_1 + \frac{3}{4})x_1 - \frac{1}{2}x_1^2$ is maximised on $[0, 1]$ at $x_1 = 1$ iff $\lambda_1 \geq \frac{1}{4}$, with value $W_\lambda(x^*) = \lambda_1 + \lambda_2 + \frac{3}{4}$. For $x_2 = 0$, $W_\lambda = 1 + (\lambda_1 + \frac{1}{4})x_1 - \frac{1}{2}x_1^2$ has maximum $1 + \frac{1}{2}(\lambda_1 + \frac{1}{4})^2$ if $\lambda_1 \leq \frac{3}{4}$ (at $x_1 = \lambda_1 + \frac{1}{4}$) and $\lambda_1 + \frac{3}{4} \leq W_\lambda(x^*)$ otherwise. Hence $\lambda \in \Lambda(x^*)$ iff $\lambda_1 \geq \frac{1}{4}$ and $\lambda_1 + \lambda_2 + \frac{3}{4} \geq 1 + \frac{1}{2}(\lambda_1 + \frac{1}{4})^2$, i.e.

$$\Lambda(x^*) \cap \{\lambda_3 = 1\} = \{\lambda_1 \geq \frac{1}{4}, \lambda_2 \geq \frac{1}{2}(\frac{3}{4} - \lambda_1)_+^2\}.$$

Its boundary contains the parabolic arc $\lambda_2 = \frac{1}{2}(\frac{3}{4} - \lambda_1)^2$, $\frac{1}{4} \leq \lambda_1 \leq \frac{3}{4}$, which no finite union of segments contains (a line meets a parabola in at most two points); since the slice of a polyhedral cone by a hyperplane is a polyhedron, $\Lambda(x^*)$ is *not polyhedral*, and $\Lambda(x^*) \subsetneq \Lambda_1(x^*)$ with a curved boundary. For instance $\lambda = (0.3, 0, 1) \in \Lambda_1 \setminus \Lambda$: the point $(0.55, 0, 1)$ improves W_λ by 0.10125. The degeneracy $F \equiv \text{const}$ is inessential. Subtract $\frac{\varepsilon}{2}x_i^2$ from each u_i , $0 < \varepsilon < 1$: $F(x) = \varepsilon x - 1$ is ε -strongly monotone, x^* is unchanged, $\Lambda_1 \cap \{\lambda_3 = 1\} = \{\lambda_1 \geq \frac{1}{4(1-\varepsilon)}\}$, and for $\frac{1}{4(1-\varepsilon)} \leq \lambda_1 \leq \frac{3}{4(1-\varepsilon)}$ a point of the slice $\lambda_3 = 1$ lies in $\Lambda(x^*)$ iff

$$\lambda_2 \geq \varphi(\lambda_1) = \frac{(\lambda_1 + \frac{1}{4})^2 / (2(1 + \varepsilon\lambda_1)) + \frac{1}{4}}{1 - \varepsilon/2} - \lambda_1, \quad \varphi''(\lambda_1) = \frac{(1 - \varepsilon/4)^2}{(1 - \varepsilon/2)(1 + \varepsilon\lambda_1)^3} > 0,$$

which reduces to the parabola for $\varepsilon = 0$; so $\Lambda(x^*)$ is not polyhedral for a strongly monotone F either. Indeed $x_3 = 1$ as before; on the edge $x_2 = 0$ the maximum, at $x_1 = \frac{\lambda_1 + 1/4}{1 + \varepsilon\lambda_1} \leq 1$, is $1 - \frac{\varepsilon}{2} + \frac{(\lambda_1 + 1/4)^2}{2(1 + \varepsilon\lambda_1)}$, which exceeds $W_\lambda(x^*) = (1 + \lambda_1 + \lambda_2)(1 - \frac{\varepsilon}{2}) - \frac{1}{4}$ iff $\lambda_2 < \varphi(\lambda_1)$. Conversely let $\lambda_2 \geq \varphi(\lambda_1)$. If the Hessian of W_λ in (x_1, x_2) is negative semidefinite, $\lambda \in \Lambda_1$ suffices; otherwise it is indefinite (its $(1, 1)$ entry is $-(1 + \varepsilon\lambda_1) < 0$), the maximum lies on an edge, and on the edges $x_1 = 1$, $x_2 = 1$ and $x_1 = 0$ the value is at most $W_\lambda(x^*)$ (increasing in x_2 ; increasing in x_1 because $\lambda_1 \geq \frac{1}{4(1-\varepsilon)}$; at most $(1 + \lambda_2)(1 - \frac{\varepsilon}{2}) < W_\lambda(x^*)$). [proved here; the counterexample was pointed out in the internal review of v0.1, the closed form for $\varepsilon > 0$ in that of v0.2]

Example 5.7 (Two commodities: $\Lambda \neq \Lambda_1$ in routing). Theorem 5.1(a),(b) use neither the Nash property nor the product structure: they hold for any finite family of C^1 functions u_k on K and any $x^* \in K$, with Λ and Λ_1 defined by the same formulas. Apply them to the costs $C_k(f) = \sum_{p \in P_k} f_p c_p(f)$ borne by the commodities of a network, $u_k = -C_k$, at a Wardrop equilibrium f^* . Take two commodities with unit demands: commodity 1 chooses between a private edge of cost $f + 2$ and a shared edge of cost f ; commodity 2 between a private edge of cost $f/4$ and the same shared edge. With $y = (f_B, f_D) \in [0, 1]^2$ the flows sent to the shared edge (path flows are determined by edge flows here, so f^* is a well-defined point),

$$C_1(y) = (1 - f_B)(3 - f_B) + f_B(f_B + f_D), \quad C_2(y) = \frac{1}{4}(1 - f_D)^2 + f_D(f_B + f_D),$$

both with indefinite Hessians (determinant -1 , from the shared-edge cross term), so (c) is unavailable. The Wardrop equilibrium is $f^* = (1, 0)$ (shared path cost $1 \leq 2$ for commodity 1, private cost $\frac{1}{4} \leq 1$ for commodity 2), unique because the reduced operator $y \mapsto (c_B - c_A, c_D - c_C)$ has Jacobian $\begin{pmatrix} 2 & 1 \\ 1 & 5/4 \end{pmatrix} \succ 0$; $C(f^*) = (1, \frac{1}{4})$. At f^* , in the reduced coordinates y (the cones Λ , Λ_1 are invariant under this affine reparametrisation of K), $N(f^*) = \mathbb{R}_+ \times \mathbb{R}_-$, $\nabla C_1(f^*) = (0, 1)$ and $\nabla C_2(f^*) = (0, \frac{1}{2})$, so $-\lambda_1 \nabla C_1 - \lambda_2 \nabla C_2 = (0, -\lambda_1 - \frac{1}{2}\lambda_2) \in N_K(f^*)$ for every $\lambda \geq 0$: $\Lambda_1(f^*) = \mathbb{R}_+^2$. For Λ , fix $\lambda_2 = 1$ ($\lambda_2 = 0$ gives $\lambda \in \Lambda$: C_1 is minimised at f^*). The Hessian of $\lambda_1 C_1 + C_2$ is $\begin{pmatrix} 4\lambda_1 & \lambda_1 + 1 \\ \lambda_1 + 1 & 5/2 \end{pmatrix}$, positive semidefinite iff $10\lambda_1 \geq (\lambda_1 + 1)^2$, in which case the first-order condition suffices and $\lambda \in \Lambda$. Otherwise the minimum over the box lies on an edge, where the restriction is a convex quadratic: on $f_B = 1$ and on $f_D = 0$ it is attained at f^* (derivatives $\lambda_1 + \frac{1}{2} + \frac{5}{2}t > 0$ and $4\lambda_1(f_B - 1) \leq 0$); on $f_D = 1$ the value is at least $\frac{15}{8}\lambda_1 + 1 > \lambda_1 + \frac{1}{4}$; on $f_B = 0$ it is $3\lambda_1 + \frac{1}{4}(1 - t)^2 + t^2$, minimised at $t = \frac{1}{5}$ with value $3\lambda_1 + \frac{1}{5}$. Hence f^* is optimal iff $\lambda_1 + \frac{1}{4} \leq 3\lambda_1 + \frac{1}{5}$, and

$$\Lambda(f^*) = \{\lambda \geq 0 : \lambda_2 \leq 40\lambda_1\} \subsetneq \Lambda_1(f^*) = \mathbb{R}_+^2.$$

For $\lambda = (0, 1)$, the flow $y = (0, \frac{1}{5})$ lowers C_2 from $\frac{1}{4}$ to $\frac{1}{5}$ (while C_1 rises from 1 to 3): f^* is a corner at which the KKT conditions hold for every weighting, while the per-commodity costs are non-convex. (f^* is also the utilitarian optimum.) [proved here; verified in E5]

Example 5.8 (Three commodities: a non-polyhedral commodity cone). Three commodities with unit demand; commodity k chooses between a private edge and one shared edge of cost f ;

the private costs are $f/4$, $f/4$ and the constant 2. With $y \in [0, 1]^3$ the flows on the shared edge and $S = y_1 + y_2 + y_3$,

$$C_k(y) = \frac{1}{4}(1 - y_k)^2 + y_k S \quad (k = 1, 2), \quad C_3(y) = 2(1 - y_3) + y_3 S.$$

The Wardrop equilibrium is $y^* = (0, 0, 1)$ (shared cost $1 \leq 2$ for commodity 3, private cost $\frac{1}{4} \leq 1$ for commodities 1, 2), unique because the reduced operator $y \mapsto (S - \frac{1}{4}(1 - y_1), S - \frac{1}{4}(1 - y_2), S - 2)$ has Jacobian $\text{diag}(\frac{1}{4}, \frac{1}{4}, 0) + \mathbf{1}\mathbf{1}^\top \succ 0$. At y^* , in reduced coordinates, $N = \mathbb{R}_- \times \mathbb{R}_- \times \mathbb{R}_+$, $\nabla C_1 = (\frac{1}{2}, 0, 0)$, $\nabla C_2 = (0, \frac{1}{2}, 0)$ and $\nabla C_3 = (1, 1, 0)$, so $\Lambda_1(y^*) = \mathbb{R}_+^3$. We claim

$$\Lambda(y^*) = \{\lambda \geq 0 : \lambda_3 + m(\lambda_1, \lambda_2) \geq 0\}, \quad m = \min_{[0,1]^2} P,$$

$P(u, w) = \frac{\lambda_1}{4}(u^2 - 2u) + \frac{\lambda_2}{4}(w^2 - 2w) + (u + w)(\lambda_1 u + \lambda_2 w)$. Write $y = (u, w, t)$, $\Phi_\lambda = \sum_k \lambda_k C_k$ and $c = (\lambda_1 + \lambda_3)u + (\lambda_2 + \lambda_3)w \geq 0$; expanding, $\Phi_\lambda(y) - \Phi_\lambda(y^*) = P(u, w) + \lambda_3 + t(c - 2\lambda_3) + \lambda_3 t^2$. For fixed (u, w) this is convex in t . If $c \geq 2\lambda_3$ its minimum over $t \in [0, 1]$ is at $t = 0$ and equals $P + \lambda_3 \geq m + \lambda_3$. Otherwise it is attained at $t = 1 - c/(2\lambda_3)$ and equals $P + c - c^2/(4\lambda_3) \geq P + \frac{c}{2} \geq P + \frac{1}{2}(\lambda_1 u + \lambda_2 w) \geq \frac{1}{4}(\lambda_1 u^2 + \lambda_2 w^2) \geq 0$. Hence y^* is a minimiser iff $\lambda_3 + m \geq 0$, and when $\lambda_3 + m < 0$ the minimiser of P on the face $t = 0$ beats it. On the slice $\lambda_2 = 1$, for $\frac{2}{3} < \lambda_1 < \frac{3}{2}$, P is strictly convex with minimiser $(3\lambda_1 - 2, \lambda_1(3 - 2\lambda_1))/((4 - \lambda_1)(4\lambda_1 - 1))$ in the open square, and $m = -\lambda_1(\lambda_1 + 1)/(4(4 - \lambda_1)(4\lambda_1 - 1))$. So the boundary of $\Lambda(y^*) \cap \{\lambda_2 = 1\}$ over that interval is the graph of the rational, non-affine function $\psi(\lambda_1) = \lambda_1(\lambda_1 + 1)/(4(4 - \lambda_1)(4\lambda_1 - 1))$: $\Lambda(y^*)$ is not polyhedral, and $\Lambda \subsetneq \Lambda_1 = \mathbb{R}_+^3$. For instance $\psi(1) = \frac{1}{18}$, and for $\lambda = (1, 1, \frac{1}{20})$ the flow $y = (\frac{1}{9}, \frac{1}{9}, 0)$ lowers Φ_λ by $\frac{1}{180}$. [proved here; the three-commodity expectation of v0.2 was confirmed numerically in the internal review, round 2; verified in E7]

Proposition 5.9 (Deciding membership is co-NP-complete). *For games with scalar actions in $[0, 1]$ and quadratic payoffs with rational coefficients, given with a Nash equilibrium x^* and a rational $\lambda \geq 0$, deciding whether $\lambda \in \Lambda(x^*)$ is co-NP-complete, and so is deciding whether $\Lambda(x^*) = \Lambda_1(x^*)$. Both remain co-NP-hard when F is strongly monotone, x^* is a vertex of K , λ is a unit vector and $\Lambda_1(x^*) = \mathbb{R}_+^N$. [proved here; pointed out in the internal review, round 2; checked by brute force in E6]*

Proof. Membership in co-NP. $\lambda \notin \Lambda(x^*)$ iff $W_\lambda(x) > W_\lambda(x^*)$ for some $x \in K$. Let $\hat{x}$ maximise W_λ over K and let Φ be the face of K in whose relative interior it lies. The restriction of the quadratic W_λ to the affine hull of Φ has a local, hence global, maximum at $\hat{x}$; it is therefore concave there, and every point of the nonempty polytope $\{x \in \Phi : \nabla_\Phi W_\lambda(x) = 0\}$ (gradient in the free coordinates) is a maximiser. A vertex of this polytope has encoding size polynomial in the input [30, Ch. 10] and is the certificate, checked in exact arithmetic (for quadratic programming over general polyhedra this is the theorem of Vavasis [32]; deciding even local optimality in an indefinite quadratic programme is NP-hard [21]). For $\Lambda \neq \Lambda_1$, a certificate is an extreme-ray generator λ of the polyhedral cone $\Lambda_1(x^*)$, of polynomial size, together with such an x for it: if every generator lay in the convex cone Λ , then $\Lambda_1 \subseteq \Lambda$. *Hardness.* We reduce from MAX-CUT [16]: given a graph with nonnegative integer weights w_{ij} on n nodes and an integer $k \geq 1$, is there a cut of weight $\geq k$? Let $\text{cut}(x) = \sum_{ij} w_{ij}(x_i + x_j - 2x_i x_j)$, $L(x) = \sum_i \deg_w(i)x_i \geq \text{cut}(x)$ on $[0, 1]^n$, and $q(x, y) = \text{cut}(x) + y(k - \frac{1}{2} - L(x))$. As q is affine in each variable separately, its maximum over $[0, 1]^{n+1}$ is attained at a vertex, where $q = \text{cut}(x)$ if $y = 0$ and $q = \text{cut}(x) - L(x) + k - \frac{1}{2} \leq k - \frac{1}{2} = q(0, 1)$ if $y = 1$; so $(0, 1)$ maximises q iff every cut has weight $< k$. Take players with actions $x_1, \dots, x_n, y, z \in [0, 1]$, fix $0 < \varepsilon < 1$, and let $u_i = -x_i - \frac{\varepsilon}{2}x_i^2$, $u_{n+1} = y - \frac{\varepsilon}{2}y^2$, $u_{n+2} = z - \frac{\varepsilon}{2}z^2 + q(x, y)$. Each payoff is concave in the own action, $F(v) = \varepsilon v + (\mathbf{1}_n, -1, -1)$ is ε -strongly monotone, and $x^* = (0, \dots, 0, 1, 1)$ is the unique equilibrium (strictly dominant strategies). $W_{e_{n+2}} = u_{n+2}$ separates in z , so $e_{n+2} \in \Lambda(x^*)$ iff every cut has weight $< k$. At x^* , $\nabla_x q = 0$ and $\partial_y q = k - \frac{1}{2} > 0$, so $\Lambda_1(x^*) = \mathbb{R}_+^{n+2}$ by (2); the other unit vectors lie in $\Lambda(x^*)$ by dominance, so $\Lambda(x^*) = \Lambda_1(x^*)$ iff $e_{n+2} \in \Lambda(x^*)$. The

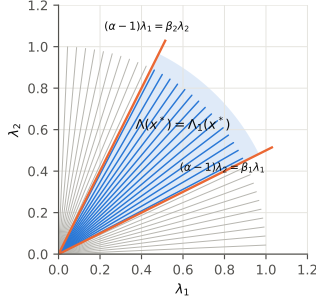


Figure 1: Example 5.4, $\alpha = 2$, $\beta_1 = \beta_2 = 0.5$: shaded, the closed-form wedge; orange, its boundaries (26.57° , 63.43°); every fifth of the 181 tested directions, blue if the exact maximisation of W_λ returns x^* , grey otherwise.

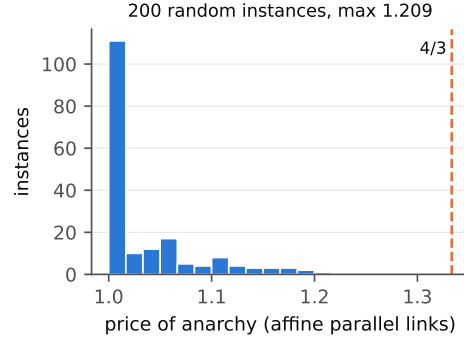


Figure 2: E1: price of anarchy of 200 random affine parallel-link instances; dashed, the bound $4/3$ of Theorem 3.7.

construction is polynomial. (Merging the first $n + 1$ players into one player with a vector action gives the same conclusions with $N = 2$.) $\square$

Remark 5.10 (What remains of Question 8.1 of v0.2). For polynomial payoffs on a polyhedral K , $\Lambda(x^*)$ is semialgebraic (Tarski–Seidenberg, 4), polyhedral for $N = 2$ (as is every closed convex cone of $\mathbb{R}^2$) and not in general for $N \geq 3$ (Example 5.6). By Proposition 5.9, unless $P = NP$ there is no polynomial-time test of $\lambda \in \Lambda(x^*)$ or of $\Lambda = \Lambda_1$, hence no polynomial-time computable description of $\Lambda(x^*)$ by polynomial inequalities. We withdraw the request of v0.2 for a characterisation “in terms of the active-face pattern and the Hessians”: in that generality it has no efficiently checkable answer. Tractable subclasses (scalar actions with N fixed; supermodular W_λ , as in Example 5.6, where polyhedrality still fails and the cut-based reduction does not apply) are directions, not questions we pose.

6 Numerical checks

All equilibria are computed by the projected-gradient iteration of Theorem 3.4(b) (Remark 3.5), run to a fixed-point step below 10^{-12} , with the natural residual $\|x - \Pi_K(x - F(x))\|$ reported; Λ_1 is tested by its linear system and Λ by exact maximisation over the box (enumeration of its 3^N faces, Appendix A). Each script states its protocol and pass/fail criteria in its docstring and writes the flags into its JSON output, from which the phrases “all criteria hold” are generated; there is no external pre-registration record (the criteria were written in the same session as the code, before the reference run). The experiments verify statements proved above; they discover nothing.

E1: routing. all 4 criteria hold. Pigou: PoA = 1.333333 for $d = 1$ and 4.7268 for $d = 16$; closed form and grid search agree to 1.5×10^{-11} . Affine parallel links: maximum PoA 1.2094 over 200 instances, 0 violations of $4/3$ (Figure 2); 86 instances have PoA = 1 exactly (e.g. when every $b_e = 0$); maximum residual 3.4×10^{-11} . The equilibrium and the optimum computed by the solver match exact water-filling to 8.8×10^{-11} . For affine costs the optimum is also the equilibrium of the tolled operator $c + \tau$ of Corollary 4.2, which is the same function as ∇C , so this is the only check of the tolls we offer (the separate “E1c” of v0.2 repeated it).

E2–E4: welfare-weight cones. all 13 criteria hold. (E2) Example 5.4: closed form, linear test and exact maximisation agree on 181/181 directions (Figure 1). Cournot (i): 0/61 grid

N	conc.	games	$\Lambda_1 \neq \{0\}$	dim Λ_1 : games	cone pts $\in \Lambda$	grid $\in \Lambda_1$: $\in \Lambda$	grid $\notin \Lambda_1$: $\notin \Lambda$	witn.
3	yes	100	28	1:4, 2:5, 3:19	115/115	575/575	8525/8525	0
4	yes	100	33	1:2, 2:7, 3:15, 4:9	153/153	307/307	16193/16193	0
3	no	100	57	1:13, 2:14, 3:30	162/206	1025/1063	8037/8037	23
4	no	100	46	1:2, 2:2, 3:13, 4:29	152/246	1265/1337	15163/15163	35

Table 1: E3–E4. Random monotone quadratic games on $[0, 1]^N$. “conc.” is joint concavity; “cone pts” are vertices and barycentre of $\Lambda_1 \cap \{\mathbf{1}^\top \lambda = 1\}$; “witn.” counts games with a certified $\lambda \in \Lambda_1 \setminus \Lambda$ among the cone points (a lower bound).

directions in Λ ; (ii): agreement 61/61. Nikaidô–Isoda modification: $\max W'_\lambda \leq 2.8 \times 10^{-17}$ on a grid of K , for all tested λ . (E2c) Example 5.6: exact test and closed form disagree on 0 of 7381 slice points; the bisected boundary is within 1.0×10^{-8} of the parabola; witness gap 0.10125. In the variant with $\varepsilon = 0.2$ the bisected boundary at 9 values $\lambda_1 \in [0.35, 0.75]$ is within 1.1×10^{-8} of φ (the tolerance of the exact test is 10^{-8}). (E3–E4, Table 1) In the 200 concave games, 61 have a non-trivial cone, and every tested point agrees with $\Lambda = \Lambda_1$ (268/268 cone points, 882/882 in-cone and 24718/24718 out-of-cone grid points), as Theorem 5.1(c) predicts. In the 200 non-concave games the inclusion $\Lambda \subseteq \Lambda_1$ is never violated, and 58 of the 103 games with a non-trivial cone carry a certified witness of strict inclusion (e.g. in game N3 nonconcave, $x^* = (1.000, 1.000, 1.000)$, $\lambda = (0.000, 0.552, 0.448) \in \Lambda_1$, the point $(0.000, 1.000, 1.000)$ improves W_λ by 0.215); witnesses are sought only among the cone points, so this is a lower bound (the 110 in-cone grid points that fail the exact test are not counted).

E5: two commodities. all 5 criteria hold. Example 5.7: $f^* = (1, 0)$ (residual 0); the bisected boundary on $\lambda_2 = 1$ is $\lambda_1 = 0.025000$ (1/40); the witness gap for $\lambda = (0, 1)$ is 0.0500. In 200 random instances, Λ_1 (computed by linear programming) is two-dimensional in 16 (the orthant in 9) and a ray in 75; 21 instances carry a certified witness of $\Lambda \neq \Lambda_1$, in 8 of them at a direction with both weights positive and in 13 only at an axis direction (a weighting that ignores one commodity). None of the 62 instances with an interior equilibrium carries one, but 4 of the 27 instances with every $b_e = 0$ (where f^* is a system optimum) do. No direction certified in Λ lies outside Λ_1 (0 violations).

E6: the reduction of Proposition 5.9. all 5 criteria hold. On all graphs with unit weights and 3–5 nodes and on random graphs with 6 nodes (unit weights, and weights in $\{1, 2, 3\}$), 1696 graphs and 13686 pairs (G, k) with $\varepsilon = 0.2$: $e_{n+2} \in \Lambda(x^*)$, decided by exact maximisation, holds iff the maximum cut, found by enumeration, is $< k$ (0 mismatches; 3598 members and 10088 non-members), and $\Lambda = \Lambda_1 = \mathbb{R}_+^{n+2}$ exactly in those cases.

E7: three commodities. all 5 criteria hold. Example 5.8: the solver returns $y^* = (0, 0, 1)$ (residual 0) and every tested direction lies in Λ_1 ; the bisected boundary on $\lambda_2 = 1$ at 16 values $\lambda_1 \in [0.7, 1.45]$ is within 1.0×10^{-8} of ψ ; the description $\{\lambda_3 + m \geq 0\}$, with m by face enumeration, agrees with the exact three-dimensional test at all 1878 grid points away from the boundary (0 mismatches); the witness gap is 0.0055555556 (1/180).

7 What can be claimed, limitations and open question

Every statement carries its status in line; the claim-by-claim table of v0.2 is now **results/CLAIMS.md**. In three lines: the classical results are theorems, proved above; the “formal relationship” between equilibrium and welfare is the first-order coincidence of Proposition 4.1, a definition rather than a mechanism; and (F, K) carries no welfare information (Proposition 5.3), so welfare

statements need the externalities as separate data, and even with them deciding whether a weighting makes an equilibrium optimal is co-NP-complete (Proposition 5.9).

Limitations. (1) Everything is finite-dimensional, deterministic and C^1 ; mixed strategies, atomic congestion games, dynamics and stochastic demand are outside the note. (2) Theorem 5.1 is a first-order statement; its only general sufficient condition for equality is concavity, which excludes Cournot-type payoffs and multi-commodity routing costs. (3) The experiments verify theorems on quadratic games over boxes and on small networks; they are exact only because $N \leq 4$ (E2–E5, E7) or $n \leq 6$ (E6), and the reported frequencies describe one sampling distribution. (4) We have not surveyed the toll-pricing and multi-objective literature exhaustively (Remark 5.2).

Open question 7.1 (Complexity for routing). In a multi-commodity network with affine costs let $C_k(f) = \sum_{p \in P_k} f_p c_p(f)$ and let f^* be a Wardrop equilibrium *in path flows* ($\Lambda(f^*)$ depends on the representative when path flows are not unique). Theorem 5.1(a),(b) give $\Lambda(f^*) \subseteq \Lambda_1(f^*)$. When no edge is shared the problem separates by commodity, each C_k is convex in the flows of commodity k , and $\Lambda = \Lambda_1$ is the face of the orthant on which $\lambda_k > 0$ only if f^{*k} minimises C_k ; otherwise $\Lambda \neq \Lambda_1$ already for two commodities (Example 5.7; E5 also with every $b_e = 0$), and Λ need not be polyhedral for three (Example 5.8). Membership $\lambda \in \Lambda(f^*)$ is in co-NP by the argument of Proposition 5.9. Is it co-NP-hard when the network and the number of commodities are part of the input? Proposition 5.9 does not apply, because the C_k have the special form above; a reduction or a polynomial algorithm would settle the question.

Next steps. Settle Question 7.1; run E4 and E5 with a global solver for larger N and m ; and confirm with the author whether the “formulation reviewed” in the line’s record was an instance of Proposition 4.1 or of Corollary 4.2, which is our reading.

A Reproducibility

The directory contains `experiments/vi_solver.py` (projections and the iteration of Theorem 3.4(b)), `traffic_poa.py` (E1), `welfare_cone.py` (E2–E4), `commodity_cone.py` (E5), `hardness_maxcut.py` (E6) and `three_commodity.py` (E7). Seed 20260930, consumed sequentially by one generator per script; Python 3.11.15, NumPy 2.4.6 [13], SciPy 1.17.1 [33] (`linprog`/HiGHS for the geometry of Λ_1); E1 took 2 s, E2–E4 114 s, E5 7 s, E6 85 s and E7 2 s on shared cores. The scripts write `results/results_*.json` (with a `criteria` block) and `results/tables_*.md`; `experiments/make_numbers.py` turns them into the macros and table body used here, so every number in the text is generated.

Solver and exact tests. Projection onto the box is clipping, onto the simplex the sort-based exact algorithm; $\gamma = \mu/L^2$ with μ the least eigenvalue of the symmetric part of the Jacobian and L its spectral norm. For each of the 3^N patterns in $\{0, 1, \text{free}\}^N$ the free coordinates solve $H_{ff}x_f = h_f - H_{fb}x_b$; candidates outside $[-10^{-9}, 1 + 10^{-9}]$ are discarded; the best candidate is the maximum over the box, because a maximiser lies in the relative interior of some face and is stationary for the restriction (if H_{ff} is singular, the same value is attained on a lower-dimensional face, also enumerated). A coordinate within 10^{-9} of a bound is active; $\lambda \in \Lambda$ iff $W_\lambda(x^*) \geq \max_K W_\lambda - 10^{-8}$; a *certified witness* improves W_λ by more than 10^{-6} .

Random families. E3–E4: $u_i(x) = -\frac{1}{2}x^\top H_i x + h_i^\top x$ on $[0, 1]^N$, concave family $H_i = B_i B_i^\top$ with Gaussian B_i , non-concave family H_i Gaussian symmetric, diagonals raised so that the symmetric part of the Jacobian of F has least eigenvalue $\frac{1}{2}$, $h_i \sim N(1, 1.5^2)$ (64–75% of equilibrium coordinates on the boundary); cone points are up to 12 vertices of $\Lambda_1 \cap \{\mathbf{1}^\top \lambda = 1\}$ and their barycentre; the grid has step $1/12$ ($N = 3$) or $1/8$ ($N = 4$); exact maximisation cross-checked

against multistart projected ascent on 13 concave instances (discrepancy 1.4×10^{-17}). E5: $a \sim U(0.05, 1)$, $b = 0$ with probability $\frac{1}{2}$ and otherwise $U(0, 1)$, 91 directions $(\cos \theta, \sin \theta)$; 3 directions accepted by the exact test only within its tolerance while failing the linear test (largest first-order violation 5.1×10^{-5}) are recorded separately. The criteria themselves are listed in the script docstrings and in `results/tables_*.md`.

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